Machine Learning in Python

Lecture 1: Course organisation and what machine learning is

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Plan for today

1. Course organisation
2. What is machine learning?
3. A short history
4. Types of machine learning

Course organisation

About the course

Format

- 7 meetings × 90 min
- short lecture + live coding in Jupyter / Colab
- notebooks published after each meeting

Tools

- Python 3, NumPy, Matplotlib
- scikit-learn (classical ML)
- Keras 3 (deep learning; JAX / TensorFlow / PyTorch backends)

Contact

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Assessment

- [e.g. project / test / attendance]
- [weights and pass threshold]
- [deadlines]

Schedule (tentative)

#	Topic	Book chapter
1	Organisation, what is ML, the perceptron	Ch. 1
2	Mathematical building blocks: tensors, gradient descent, backprop	Ch. 2
3	Frameworks: TensorFlow, PyTorch, JAX and Keras; first MLP	Ch. 3
4	Classification and regression with dense networks	Ch. 4
5	Generalisation, overfitting, evaluation, regularisation	Ch. 5–6
6	Image classification with convolutional networks	Ch. 8
7	[project presentations / test]	—

Chapters refer to F. Chollet, M. Watson, *Deep Learning with Python*, 3rd ed.

Literature

Main textbook

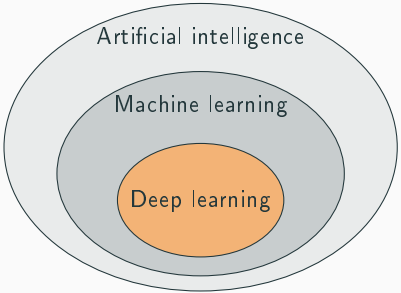
- F. Chollet, M. Watson, *Deep Learning with Python*, 3rd ed., Manning, 2025.
  Full text online: `https://deeplearningwithpython.io`

Supplementary

- A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, 3rd ed., O'Reilly, 2022.
- G. James, D. Witten, T. Hastie, R. Tibshirani, J. Taylor, *An Introduction to Statistical Learning with Applications in Python* (ISLP), Springer, 2023. Free PDF: `https://www.statlearning.com`
- S. J. D. Prince, *Understanding Deep Learning*, MIT Press, 2023. Free PDF: `https://udlbook.github.io/udlbook`

What is machine learning?

AI, machine learning, deep learning



- **AI**: automating intellectual tasks normally performed by humans. Includes hand-written rule systems ("symbolic AI").
- **ML**: systems whose behaviour is learned from data rather than written explicitly.
- **DL**: ML with models built from many layers of learned representations (neural networks).

Two classic definitions

Informal (usually attributed to A. Samuel, 1959)

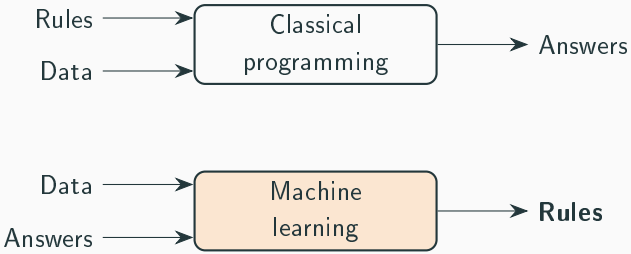
Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

Operational (T. Mitchell, *Machine Learning*, 1997)

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

Example (spam filter): T = label emails as spam / not spam, E = archive of labelled emails, P = fraction of correctly labelled new emails.

A new programming paradigm



A machine-learning system is *trained* rather than explicitly programmed: it is shown many examples and finds statistical structure that lets it map inputs to outputs. (after Chollet & Watson, Ch. 1)

Vocabulary we will use throughout the course

sample one example x_j

features coordinates of $\mathrm{x}_j \in \mathbb{R}^n$

label / target desired output d_j

model function f_w with parameters w

loss measure of how wrong $f_\mathrm{w}(\mathrm{x}_j)$ is

training adjusting w to reduce the loss on training data

test set data *not* used in training

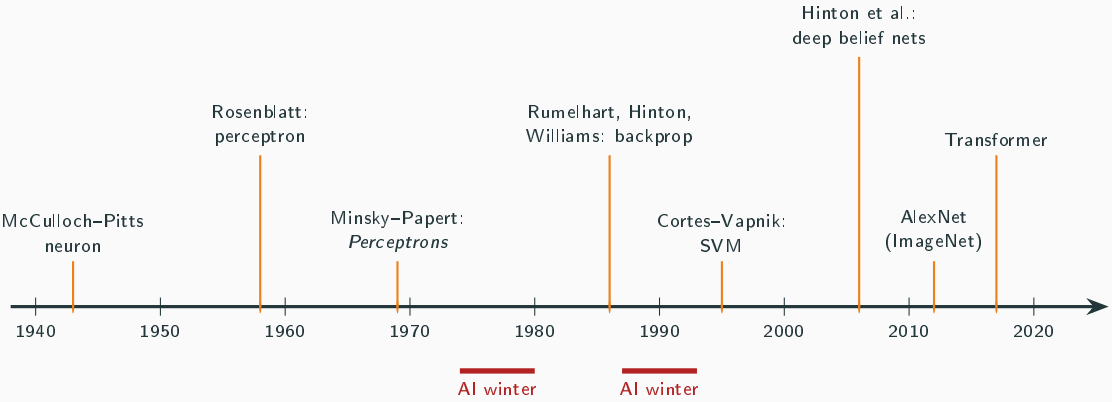
generalisation performance on unseen data

Key point

The goal is never to fit the training data per se, but to perform well on *new* data.

A short history

Milestones (selection)



The boundaries of the "AI winters" are approximate and refer to AI as a whole, not only to neural networks.

Where neural networks come from

- **1943**: W. McCulloch & W. Pitts — a logical model of a neuron: a threshold unit with fixed weights.
- **1958**: F. Rosenblatt — the perceptron: a threshold unit whose *weights are learned* from examples (topic of the second part of today's lecture).
- **1960**: B. Widrow & M. Hoff — ADALINE, trained by minimising squared error (the "delta rule").
- **1969**: M. Minsky & S. Papert, *Perceptrons* — rigorous analysis of what single-layer perceptrons cannot compute (e.g. XOR / parity). Interest and funding in neural networks declined.
- **1986**: D. Rumelhart, G. Hinton, R. Williams — popularised backpropagation for training multilayer networks (the method itself has earlier roots: Linnainmaa 1970, Werbos 1974).

From “shallow” to deep learning

- **1990s–2000s**: kernel methods (SVM, Cortes & Vapnik 1995), decision trees, random forests, gradient boosting dominate practical ML.
- **1989–1998**: Y. LeCun et al. — convolutional networks for handwritten digits (LeNet).
- **2006**: G. Hinton, S. Osindero, Y.-W. Teh — greedy layer-wise pretraining of deep belief nets; revived interest in deep architectures.
- **2012**: A. Krizhevsky, I. Sutskever, G. Hinton — AlexNet, a GPU-trained ConvNet, wins ImageNet (ILSVRC) with top-5 error $\approx 15\%$ vs. $\approx 26\%$ for the runner-up.
- **2017**: Vaswani et al. — the Transformer; basis of today’s large language models.
- **2018**: Turing Award for Y. Bengio, G. Hinton, Y. LeCun.

Why now? Data (internet scale), hardware (GPUs/TPUs), algorithms (better activations, initialisation, optimisers, normalisation).

Types of machine learning

Overview

Paradigm	What the data look like	Typical tasks
Supervised	pairs $(\mathbf{x}_j, d_j)$	classification, regression
Unsupervised	only $\mathbf{x}_j$	clustering, dimensionality reduction, density estimation
Self-supervised	only $\mathbf{x}_j$; targets built from the input itself	next-word prediction, masked image modelling
Semi-supervised	few labelled, many unlabelled	classification with scarce labels
Reinforcement	states, actions, rewards	games, control, robotics

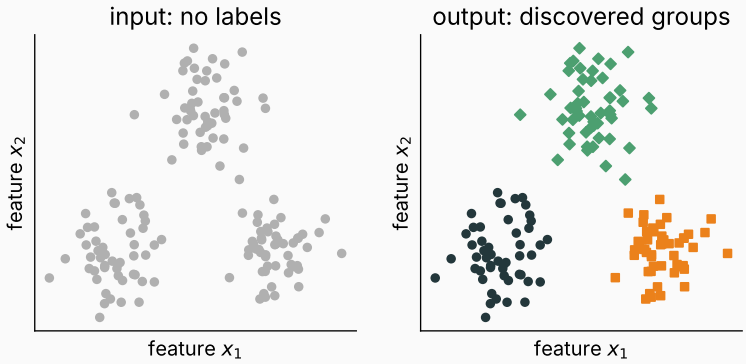
Supervised learning

- Training data: $\{(\mathrm{x}_j, d_j)\}_{j=1}^{s}$, each input comes with the correct answer.
- Goal: learn f with $f(\mathrm{x}) \approx d$ *also for new* x.
- **Classification**: d is a category (spam / not spam, digit 0–9).
- **Regression**: d is a number (house price, demand).



Unsupervised learning

- Only inputs $\{x_j\}$, no labels. Goal: discover structure in the data.
- Clustering (customer segmentation), dimensionality reduction / visualisation (PCA, t-SNE, UMAP), anomaly detection.
- No single "correct answer" $\Rightarrow$ evaluation is harder and often task-dependent.



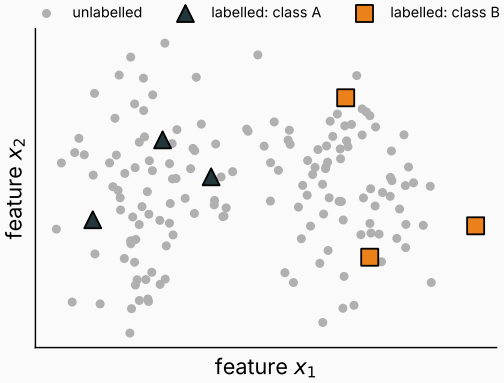
Self-supervised and semi-supervised learning

Self-supervised

- Supervised learning *without human labels*: targets are generated from the input itself.
- Example: predict the next word of a text; reconstruct a masked part of an image.
- This is how large language models are pretrained.

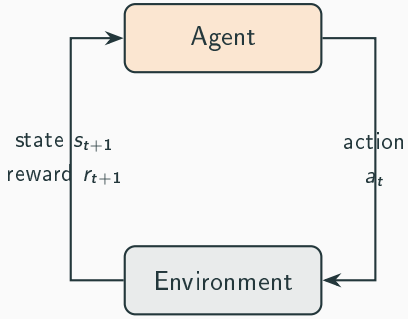
Semi-supervised

- A few labelled points, many unlabelled.
- Unlabelled data reveal the shape of the classes.



Reinforcement learning

- An **agent** observes a **state**, takes an **action**, receives a **reward** and a new state.
- Goal: a **policy** (state $\mapsto$ action) maximising expected cumulative reward.
- Feedback is delayed and evaluative, not "the correct answer".
- Formal framework: Markov decision processes; closely related to dynamic programming.
- Examples: AlphaGo, robot control.



This course

- Mostly supervised learning with neural networks.
- Today: the simplest trainable neuron — the **perceptron**.
- Next lectures: why we need many neurons in layers (MLP), how to train them with **gradient descent**, and how to do it in Keras.